

CYBERSECURITY INCIDENT REPORT

Failed Privilege Escalation Attempt Detection

Project: Linux Server Hardening and Security Monitoring Lab

Environment: Ubuntu Server Virtual Machine

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This report documents the detection, analysis, and investigation of a failed authentication event identified through real-time security monitoring.

Executive Summary

During routine security monitoring activities, a failed privilege escalation attempt was detected on the Ubuntu Server laboratory environment. The event occurred when an attempt was made to access the root account using invalid credentials. The security monitoring process successfully identified the event through authentication log analysis. No unauthorized access was obtained and no systems were compromised.

Incident Overview

Incident Type: Failed Authentication Attempt

Classification: Privilege Escalation Attempt

Severity: Low

Status: Closed / Resolved

Affected Asset: Ubuntu-SIEM Virtual Machine

Target Account: root

Environment Description

The monitored environment consisted of an Ubuntu Server virtual machine deployed in Oracle VirtualBox. Security controls implemented included UFW Firewall, Fail2Ban, OpenSSH, and system authentication logging. Real-time monitoring was performed using Linux command-line tools.

Detection Methodology

Authentication logs were monitored using the command 'sudo tail -f /var/log/auth.log'. This allowed real-time visibility into login activities, authentication events, session creation, and privilege escalation attempts occurring on the system.

Evidence Analysis

Log analysis identified a FAILED LOGIN event associated with an attempt to execute the 'su root' command. PAM (Pluggable Authentication Modules) recorded an authentication failure because invalid credentials were supplied. The event confirms that access controls functioned correctly and prevented unauthorized privilege escalation.

Root Cause Analysis

The event was intentionally generated as part of a controlled security monitoring exercise. An incorrect password was entered during a root access attempt. The resulting log entries demonstrate how authentication failures are recorded and detected within Linux environments.

Impact Assessment

No security breach occurred. No unauthorized access was obtained. No sensitive data was exposed, modified, or destroyed. System availability, integrity, and confidentiality remained unaffected.

Response Actions

1. Monitored authentication logs in real time.
 2. Identified failed authentication event.
 3. Verified affected account and source terminal.
 4. Confirmed unsuccessful privilege escalation.
 5. Preserved evidence through screenshots and documentation.
 6. Documented findings in an incident report.
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Lessons Learned

Authentication logs provide valuable indicators of suspicious activity. Continuous monitoring enables rapid detection of unauthorized access attempts. Security controls such as UFW and Fail2Ban enhance defensive capabilities and should be maintained as part of a defense-in-depth strategy.

Recommendations

Continue monitoring authentication logs, implement centralized log collection, configure automated alerting for repeated authentication failures, regularly review access controls, and expand the environment into a SIEM-based monitoring platform.

Appendix A – Evidence Screenshot

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Jaspher@ubuntu-siem:~$ sudo tail -f /var/log/auth.log
2026-06-06T07:59:17.747020+00:00 ubuntu-siem agetty[1767]: tty2: Invalid character 0x1b in login name
2026-06-06T08:02:46.652583+00:00 ubuntu-siem login: pam_unix(login:auth): check pass; user unknown
2026-06-06T08:02:46.663242+00:00 ubuntu-siem login: pam_unix(login:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty3 ruser= rhost=
2026-06-06T08:02:50.203484+00:00 ubuntu-siem login: FAILED LOGIN 1 FROM tty3 FOR su root. Authentication failure
2026-06-06T08:05:33.098547+00:00 ubuntu-siem sudo: pam_unix(sudo:session): session closed for user root
2026-06-06T08:05:51.848605+00:00 ubuntu-siem sudo: pam_unix(sudo:session): session opened for user root(uid=0) by jaspher(uid=1000)
2026-06-06T08:05:51.849018+00:00 ubuntu-siem sudo: jaspher : TTY=/dev/tty1 : PWD=/home/jaspher : USER=root : COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-06-06T08:06:16.185508+00:00 ubuntu-siem sudo: pam_unix(sudo:session): session closed for user root
2026-06-06T08:06:39.908308+00:00 ubuntu-siem sudo: pam_unix(sudo:session): session opened for user root(uid=0) by jaspher(uid=1000)
2026-06-06T08:06:39.908448+00:00 ubuntu-siem sudo: jaspher : TTY=/dev/tty1 : PWD=/home/jaspher : USER=root : COMMAND=/usr/bin/tail -f /var/log/auth.log
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Figure 1. Authentication log entries showing a FAILED LOGIN event for the root account and related session activity observed during real-time monitoring.